Supplementary Materials for

Burst-selective motor coding inspires multiplicative gating for efficient reinforcement learning

Supplementary Figures

Figure S1 | Burst-fraction null-model, generality, and fragmentation controls

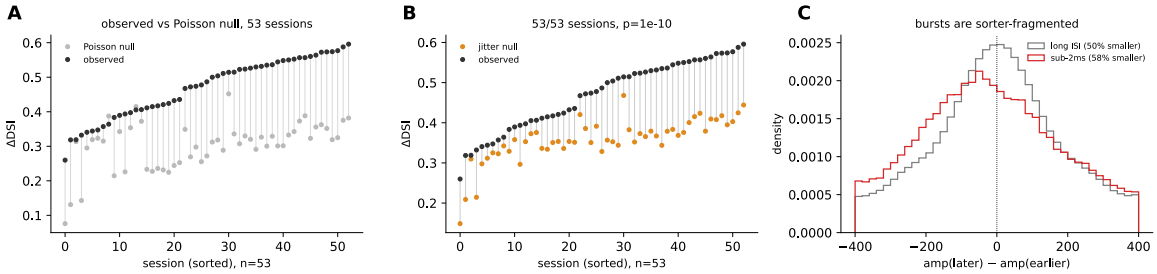


Fig. S1: Burst-fraction null-model, generality, and fragmentation controls. (A) Observed vs. Poisson-PSTH-null ΔDSI (burst − rate) for all 53 sessions (37 Indy M1, 15 monkey C, and the MC_Maze benchmark session; sorted). The Poisson null is unreliable and can approach or exceed the observed value. **(B)** Observed vs. timing-preserving jitter-null ΔDSI for the same 53 sessions (sorted); 53/53 sessions lie above the jitter null, session-level Wilcoxon $p = 1.2 \times 10^{-10}$. **(C)** Evidence consistent with spike-sorter fragmentation: amplitude difference (later − earlier) for sub-2 ms cross-unit spike pairs (red) vs. long-interval pairs (grey); the later spike is systematically smaller only at burst timescales, suggesting that per-unit burst fraction may be under-counted.

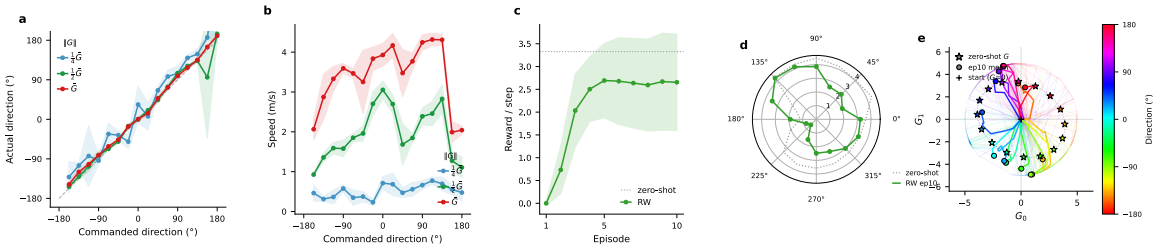


Fig. S2: Zero-shot directional control and reward-weighted cold-start adaptation. (A) Actual movement direction as a function of commanded direction for three G magnitudes ($\frac{1}{2}\bar{G}$, $\bar{G}$, $2\bar{G}$, where $\bar{G}$ is the mean zero-shot norm). Across all amplitudes, the ant tracks the commanded direction closely (dashed line: identity), demonstrating that the direction of G controls movement direction independently of its magnitude. **(B)** Locomotion speed as a function of commanded direction for the same three amplitudes. Speed is modulated by $\|G\|$ while remaining approximately uniform across directions, confirming that G magnitude and direction act as independent control axes for speed and heading. **(C)** Cold-start adaptation with reward-weighted G updates (RW). Starting from $G = 0$, the agent recovers behaviorally useful G vectors within a few episodes. Mean reward per step ($\pm$ s.d. across 16 directions) is shown; the dotted line indicates the zero-shot performance level. **(D)** Polar plot of reward per step at episode 10 as a function of movement direction, for the zero-shot policy (dotted) and after RW adaptation (solid). Adaptation improves performance across all directions. **(E)** Trajectories of the G vector in the G_0-G_1 plane during RW adaptation. Stars mark the zero-shot G positions computed by the pretrained g-layer; circles mark the mean adapted G after 10 episodes. Arrows indicate the direction of adaptation; the dotted curve shows the zero-shot locus across all commanded directions.

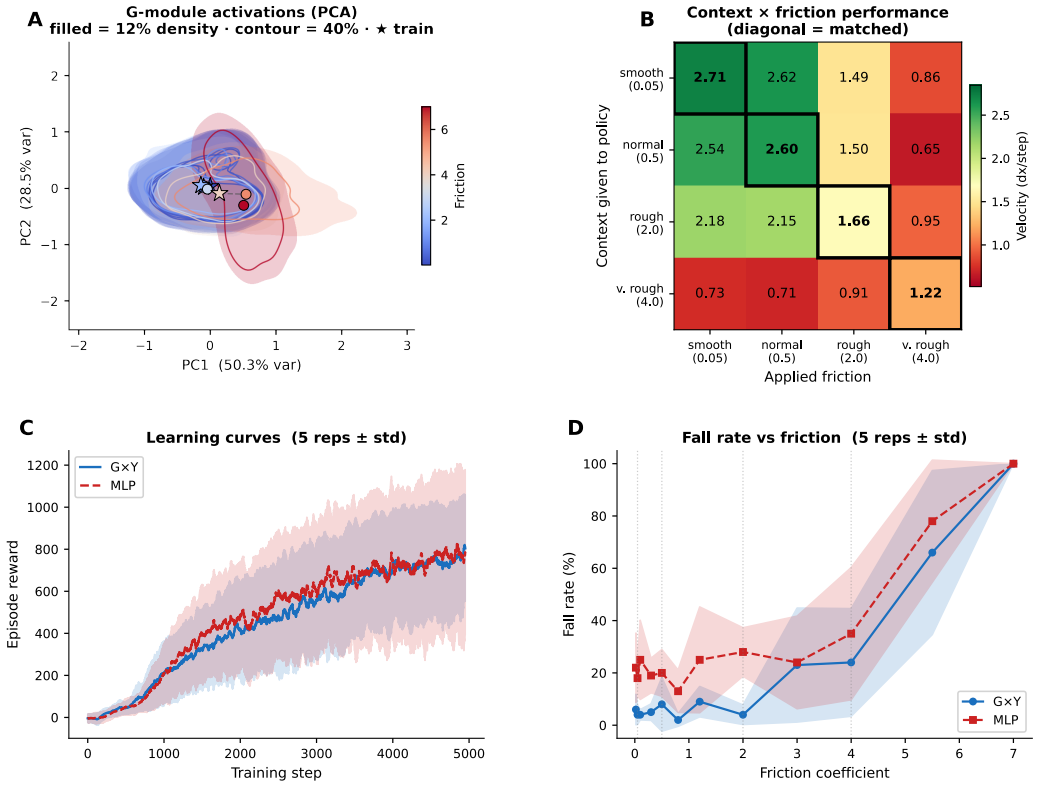


Fig. S3: Multiplicative context-gating enables robust generalisation across floor friction conditions. **(A)** Principal component analysis of the G-module activations collected over 100 evaluation episodes per friction level. Each point represents the gate vector $\mathbf{g} = G(\mathbf{s}, \mathbf{c})$ at a single timestep, coloured by friction coefficient (blue = low, red = high). Stars denote training frictions; circles denote novel test frictions. The smooth trajectory of centroids indicates that the gating network encodes friction as a continuous latent variable despite receiving only discrete one-hot context during training. **(B)** Context-friction performance matrix. Each entry shows the mean forward velocity (dx/step) when the agent is given context $\mathbf{c}_i$ (row) and deployed on friction f_j (column), averaged over 10 episodes. Bold diagonal entries correspond to matched context-friction pairs; the diagonal is consistently the highest in each row. **(C)** Training curves (mean ± std over 5 seeds). Both G×Y and the MLP baseline achieve comparable final episode reward. **(D)** Fall rate as a function of floor friction (mean ± std, 5 seeds, 20 episodes per condition). G×Y (blue) maintains near-zero fall rate up to $f = 2.0$ and degrades more gracefully than the MLP baseline (red).

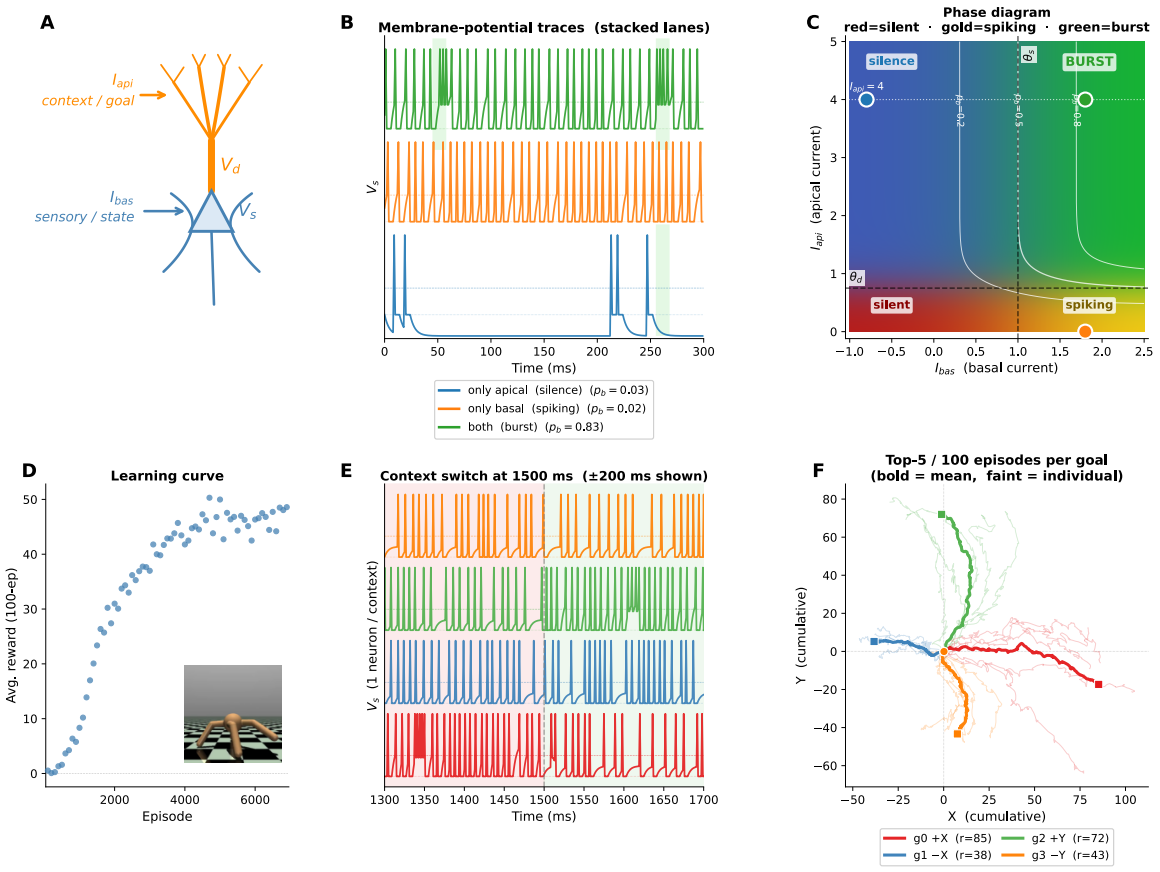


Fig. S4: Burst-gated neural actor on the four-goal MuJoCo Ant-v4 locomotion task. The Ant version of the two-compartment burst model (Fig. 2 shows the two-link reaching version). Panels A-C are identical in mechanism to Fig. 2A-C. **(A)** Two-compartment pyramidal neuron model. Feedforward basal current I_{bas} drives the somatic voltage V_s; top-down apical current $I_{\text{api}} = g \cdot I_{\text{API_SCALE}}$ ($I_{\text{API_SCALE}} = 4.0$) drives the dendritic voltage V_d. Burst requires co-activation of both compartments. **(B)** Somatic membrane-potential traces (V_s, 300 ms) under three input conditions (stacked lanes). Blue: apical only ($I_{\text{api}} = 4.0$, $I_{\text{bas}} = -0.8$): gate open but soma sub-threshold, near-silence ($p_{\text{burst}} = 0.03$). Orange: basal only ($I_{\text{bas}} = 1.8$, $I_{\text{api}} = 0$): regular spiking, burst gate closed ($p_{\text{burst}} = 0.02$). Green: both ($I_{\text{bas}} = 1.8$, $I_{\text{api}} = 4.0$): dense recurrent bursting ($p_{\text{burst}} = 0.83$, green shading marks burst windows). **(C)** Two-dimensional phase diagram of the burst mechanism using a bilinear RGB colormap encoding $p_s(I_{\text{bas}})$ and $p_d(I_{\text{api}})$ independently. Four regions are visually separated: red (silent, $p_s \approx 0$, $p_d \approx 0$), blue (apical-only silence, $p_s \approx 0$, $p_d \approx 1$), gold (spiking without burst, $p_s \approx 1$, $p_d \approx 0$), and green (burst, $p_s \approx 1$, $p_d \approx 1$). Contours show $p_{\text{burst}} \in \{0.2, 0.5, 0.8\}$. Dashed lines mark $\theta_s = 1.0$ and $\theta_d = 0.75$; dotted white line marks the active apical drive $I_{\text{api}} = 4.0$. Coloured dots correspond to the three conditions in (B). **(D)** Learning curve for the four-goal MuJoCo Ant-v4 task (100-episode moving average). The agent converges to average rewards above 30 within ∼3500 episodes. Inset: screenshot of the simulated ant. **(E)** Post-training V_s traces (stacked lanes, one neuron per goal context) showing 200 ms before and 200 ms after the context switch at $t = 1500$ ms (100 steps × 15 ms). Goal 0 (red) bursts densely in the preceding window and stops abruptly at the switch; goal 2 (green) transitions from sub-threshold silence into sustained bursting. Goals 1 and 3 remain sub-threshold throughout, confirming context selectivity. **(F)** Top-5 out of 100 evaluation episodes per goal direction (faint = individual, bold = mean; circles = start, squares = end). Mean rewards: 85 ($+X$), 38 ($-X$), 72 ($+Y$), 43 ($-Y$).

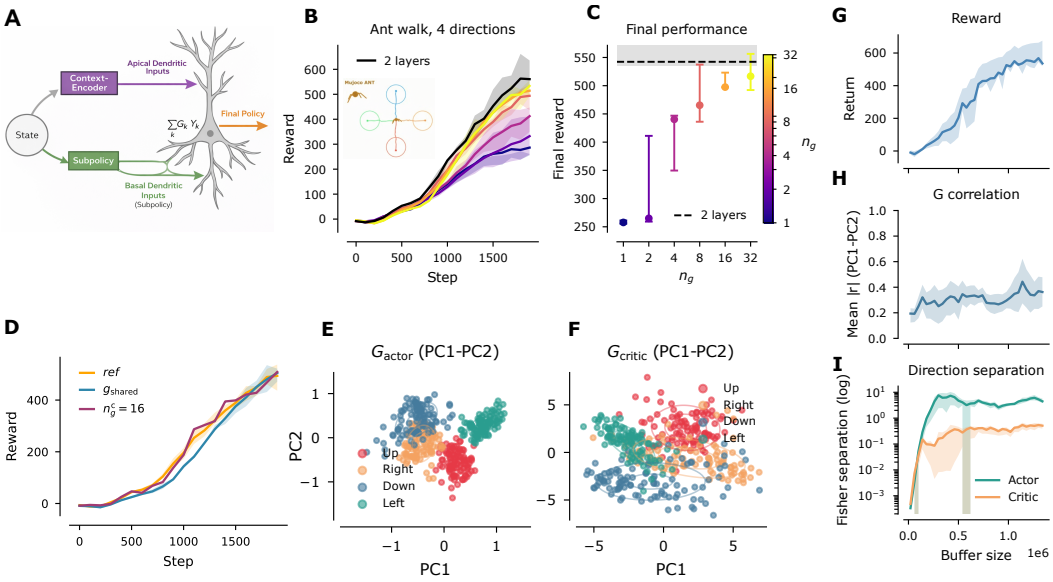


Fig. S5: Bilinear actor–critic architecture and behaviour. (A) Architecture and its biological interpretation: a context encoder provides the goal-dependent gate (apical input), a subpolicy provides state-dependent primitives (basal input), and their multiplicative combination $\sum_k G_k Y_k$ forms the policy. **(B)** Learning curves on the MuJoCo Ant directional locomotion task, comparing a two-layer nonlinear MLP with a single linear layer combined through bilinear gating, for increasing numbers of gating units n_g. **(C)** Final performance as a function of n_g; with enough gating units the single gated linear layer matches the two-layer MLP. **(D)** Learning curves for shared versus independent actor and critic gating: sharing G does not cost performance. **(E,F)** Actor and critic gating vectors projected onto their first two principal components, coloured by movement direction. **(G)** Training return. **(H)** Correlation between actor and critic gating vectors over training. **(I)** Direction separation (Fisher) of actor and critic gating over training.

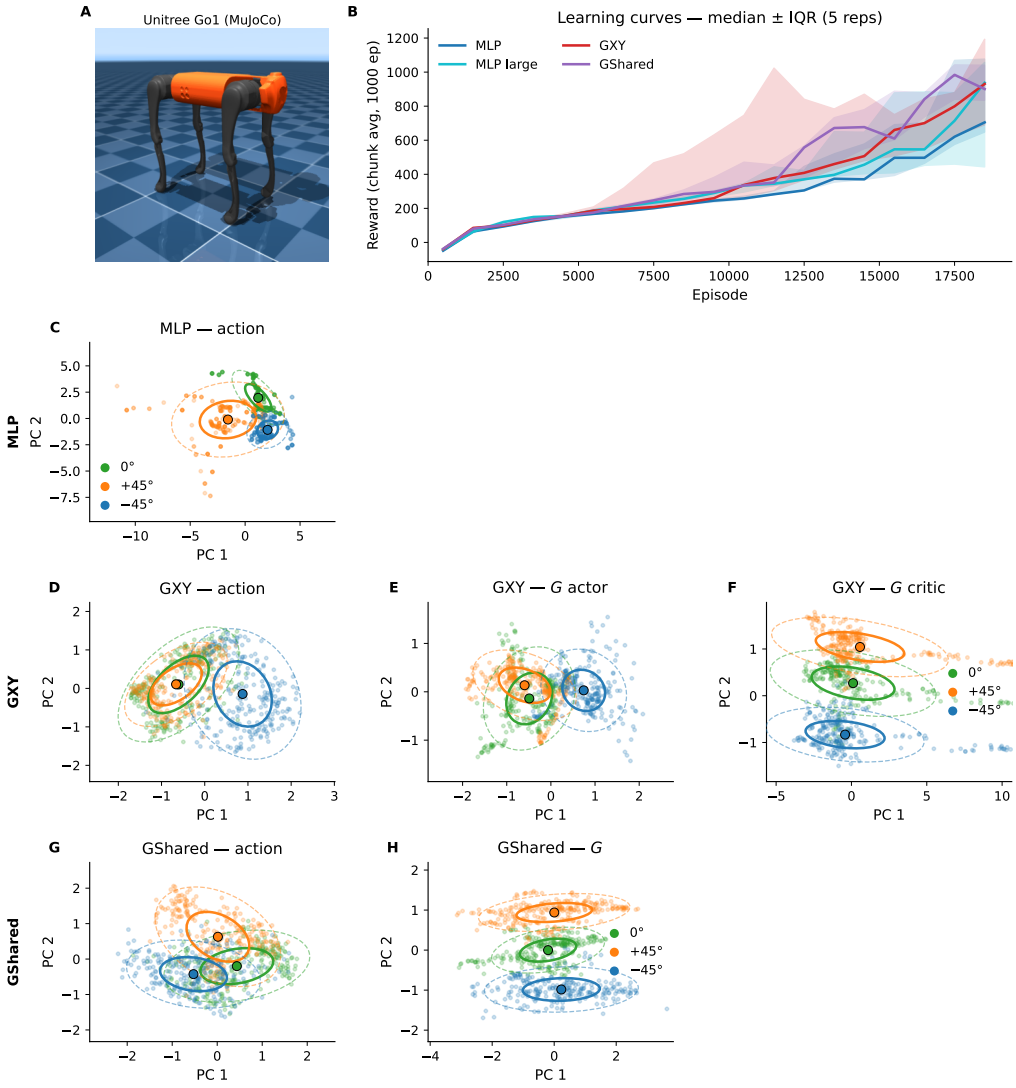


Fig. S6: Gating representations in SAC locomotion on the Unitree Go1 quadruped. **(A)** MuJoCo simulation environment. The robot is rewarded for walking in a direction $\theta \in \{-45°, 0°, +45°\}$, encoded as $[\cos\theta, \sin\theta]$ and changed randomly every 500 steps within each 1500-step episode. **(B)** Learning curves for four architectures (median ± IQR over five independent runs): MLP baseline, MLP-large (parameter-matched critic), GXY (independent gating in actor and critic), and GShared (shared G layer). **(C–H)** PC1 versus PC2 projections of representations collected across 15 rollouts per direction. Gaussian ellipses mark the 1σ and 2σ contours. **(C)** MLP action space shows overlapping distributions across directions. **(D–F)** GXY actor action space, actor gate, and critic gate; both gating spaces exhibit directional clustering. **(G,H)** GShared action space and shared gate; the gate develops direction-selective activity despite being trained only through the critic loss.